

Week 6: Prototype Development (Phase 1)

Temporal CNN for DDoS Attack Detection

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1. Goal

Implement the core AI detection pipeline and produce trained temporal models capable of classifying network flows as Benign, Attack, or Suspicious on the BCCC-Cloud-DDoS-2024 dataset. The target was $\geq 80\%$ 3-class test accuracy. The TF/Keras visual-strip baseline in `ids_2.ipynb` reaches 94.05% on binary classification but collapses to $\sim 47\%$ on 3-class due to complete Suspicious class failure. Week 6 iterated on both the model architectures and the training pipeline through two rounds of changes, culminating in full training runs on both the BiLSTM + Attention and 1D Residual CNN models.

2. Week 5 Completion Status

The following deliverables from Week 5 were confirmed complete before Week 6 work began:

Component	Details	Status
Temporal image pipeline	Flow-to-image conversion logic finalised; time window sizes (5s, 10s, 20s, 50s) tested on sample batches	Complete
End-to-end data flow	Raw BCCC flows $\rightarrow$ temporal images $\rightarrow$ CNN input format validated on sample batches	Complete
Technical barrier analysis	Memory and compute requirements profiled; mitigation strategies documented for class imbalance and input dimension variability	Complete
Prototype architecture design	Four-stage pipeline: Data Ingestion $\rightarrow$ Temporal Imaging $\rightarrow$ CNN Model $\rightarrow$ Evaluation	Complete
Baseline CNN (<code>ids_2.ipynb</code>)	Binary: 94.05% accuracy, AUC 0.976, Precision 0.95, Recall 0.88. Multi-class: $\sim 47\%$ — Suspicious recall $\sim 0\%$	Complete

3. Activities

3.1 Data Pipeline Setup

The full 700,774-flow dataset was loaded from merged_CSVs.csv on a Quadro P5000 GPU (load time ~105s) and passed through a cleaning pipeline: chronological sorting by timestamp; replacement of the non-numeric sentinel “not a complete handshake” in delta_start and handshake_duration with 0; protocol label encoding (TCP=0); dropping flow_id, src_ip, dst_ip, and timestamp as identifiers; filling 8,394,232 NaN and Inf values with 0; and dtype downcasting to reduce memory from 13,738 MB to 983 MB. The cleaned dataset (700,774 rows × 318 feature columns) was saved to ddos_clean.parquet.

Feature selection was applied in two passes, both fit on the first 490,541 rows (70% of data) to prevent test-set leakage. VarianceThreshold(0.001) reduced 318 features to 154. Correlation pruning then removed 68 redundant features where Pearson $r > 0.97$, leaving 86 features. The selectors were saved to models/ for reproducible inference. A MinMaxScaler fit on the same 70% of rows was saved as temporal_scaler.pkl; the scaled matrix spans -5.37 to 20.96 across all 700,774 rows.

Windows of 30 consecutive flows with stride 15 (50% overlap) were built over the full time-sorted dataset, yielding 46,717 windows: 15,202 Attack (32.5%), 27,571 Benign (59.0%), and 3,944 Suspicious (8.4%). Each window’s label was set to the label of its last flow rather than a majority vote. Majority voting was ruled out because Suspicious flows (8.4% of all flows) almost never win a vote in a 30-flow window, silently dropping the class from the training signal.

A stratified 70/15/15 split at the window level produced: train 32,701 windows (Attack 10,641 / Benign 19,299 / Suspicious 2,761), val 7,008 (Attack 2,280 / Benign 4,136 / Suspicious 592), test 7,008 (Attack 2,281 / Benign 4,136 / Suspicious 591). This replaced the initial plain temporal cut on raw flows, which had left only 2 Suspicious windows in training while the test set was 38% Suspicious — causing both models to ignore the class entirely.

3.2 Architecture Iteration

Two rounds of changes were made before the final training run. Each round addressed a specific failure identified from the previous configuration.

Round 1 — Architecture Improvements

Change	Before	After	Why
Feature count	318 raw features	86 features after VarianceThreshold + correlation pruning	Near-constant and correlated features produce noisy gradients; pruning retains key DDoS indicators including rst_flag_percentage_in_total, syn_flag_percentage_in_total, and packet_IAT_max
Window label	Majority vote over 30 flows	Label of the last flow	Suspicious is 8.4% of flows and almost never wins a majority vote; last-flow label preserves exact ground truth
Window size	20 flows	30 flows	Longer windows provide more temporal context for detecting gradual attack ramp-up patterns

Change	Before	After	Why
LSTM output	Last hidden state only	Soft attention over all 30 timesteps	Attention up-weights the most anomalous flows in the window rather than compressing the entire sequence into one vector
CNN blocks	Plain Conv1d layers	Residual blocks with skip connections	Skip connections prevent gradient vanishing in deeper convolutional stacks
Input layer	Raw features into LSTM / Conv1d directly	Linear($F \rightarrow 128$) + BatchNorm1d projection first	BatchNorm stabilises the wide range of magnitudes across 86 heterogeneous network statistics

Round 2 — Stratified Split and Regularisation

After Round 1, a more fundamental problem was identified: the plain temporal split left only 2 Suspicious windows in training. Both models predicted only Benign and Attack. The following changes resolved this:

Change	Before	After	Why
Split strategy	Temporal 70/15/15 cut on raw flows	Stratified 70/15/15 at window level	Guarantees all three classes appear in training, validation, and test
Scaler / selector scope	Applied to each split independently	Fit on first 70% rows; applied to all windows	Eliminates data leakage from test distribution into normalisation and feature pruning
Dropout	0.3	0.4	Stronger regularisation after overfitting observed on the corrected split
1D CNN intra-block dropout	None	Dropout(0.2) inside each residual block	Reduces feature co-adaptation within convolutional layers
BiLSTM head	Linear($256 \rightarrow 128 \rightarrow 64 \rightarrow 3$)	Linear($256 \rightarrow 64 \rightarrow 3$)	Simpler head generalises better given the existing dropout and weight decay
Weight decay	$1e-4$	$1e-3$	Stronger L2 penalty after the corrected split revealed the models were memorising training windows
Loss function	CrossEntropyLoss	CrossEntropyLoss + label smoothing 0.1	Prevents overconfident predictions on Benign (58.9%) and maintains probability mass on Suspicious (8.4%)
LR scheduler	ReduceLROnPlateau	CosineAnnealing LR	ReduceLROnPlateau drops LR sharply at plateaus, destabilising training under weight decay $1e-3$; cosine decay is smoother

Change	Before	After	Why
Learning rate	1e-3	3e-4	More stable alongside the strengthened regularisation

3.3 Model Training

Both models were trained under identical conditions on a Quadro P5000 GPU: Adam lr=3e-4, weight decay 1e-3, label smoothing 0.1, CosineAnnealingLR, batch size 256, EarlyStopping on val loss with patience 10.

BiLSTM + Attention (688,260 parameters)

Training ran for 30 epochs. Val loss improved consistently from 0.7799 at epoch 1 to a best of 0.5687 at epoch 28, with val accuracy climbing from 71.5% to 93.9%. The model converged without overfitting — train and val accuracy tracked closely across all 30 epochs.

1D Residual CNN (481,283 parameters)

Training ran to the 30-epoch early stopping limit. Val loss improved from 0.7516 at epoch 1 to a best of 0.5779 at epoch 23, with val accuracy reaching 94.1%. The 1D CNN converged faster than the BiLSTM — it crossed 90% val accuracy by epoch 6 compared to epoch 9 for the BiLSTM — and with 30% fewer parameters.

4. Milestone / Deliverable: Prototype Training Results

4.1 Test Results

Model	Params	Test Accuracy	Macro F1	Macro Recall	≥80% Target?
BiLSTM + Attention	688,260	93.84%	0.8996	0.9367	Yes
1D Residual CNN	481,283	94.11%	0.9035	0.9337	Yes
CNN Visual Strip — 3-class (reference)	—	~47%	~0.38	~0.33	No

4.2 Per-Class Results

BiLSTM + Attention:

Class	Precision	Recall	F1-Score	Support
Attack	0.9581	0.9325	0.9451	2,281

Class	Precision	Recall	F1-Score	Support
Benign	0.9772	0.9420	0.9593	4,136
Suspicious	0.6904	0.9357	0.7945	591

1D Residual CNN:

Class	Precision	Recall	F1-Score	Support
Attack	0.9450	0.9408	0.9429	2,281
Benign	0.9809	0.9449	0.9626	4,136
Suspicious	0.7185	0.9154	0.8051	591

Both models recovered the Suspicious class that the visual-strip baseline (ids_2.ipynb) could not detect at all. Suspicious recall of 93.6% (BiLSTM) and 91.5% (1D CNN) confirms that the 30-flow temporal window gives the models enough context to identify the clustered Suspicious traffic from December 18–19. Suspicious precision is lower (69.0% and 71.9% respectively), indicating some Benign and Attack windows are misclassified as Suspicious — an acceptable trade-off at this stage given the class rarity. The 1D CNN achieves a marginally higher overall accuracy and Suspicious F1 with 30% fewer parameters, making it the more efficient of the two architectures.

4.3 Saved Artifacts

File	Description
ddos_clean.parquet	Cleaned, time-sorted dataset (700,774 rows × 322 columns including label and activity)
models/bilstm_ids.pt	BiLSTM + Attention weights from best val loss checkpoint (epoch 28)
models/cnn1d_ids.pt	1D Residual CNN weights from best val loss checkpoint (epoch 23)
models/temporal_scaler.pkl	MinMaxScaler fit on first 70% of rows
models/variance_selector.pkl	VarianceThreshold(0.001) selector fit on first 70% of rows
models/corr_keep_idx.npy	Column indices retained after correlation pruning ($r > 0.97$)
models/label_encoder.pkl	LabelEncoder mapping: 0=Attack, 1=Benign, 2=Suspicious

5. Next Steps (Weeks 7–8: Baseline Benchmarking)

- Profile resource usage on the Quadro P5000: training time per epoch, inference latency per 30-flow window, and memory footprint for both models.

- Run activity-level (26-class) classification on the best-performing model (1D Residual CNN) to assess whether the temporal pipeline can resolve individual attack types beyond the 3-class label.
- Evaluate both temporal models on the binary task (Benign vs. Attack+Suspicious combined) and compare against the 94.05% single-flow baseline to quantify the benefit of temporal context for binary detection.
- Generate confusion matrices for both models and analyse the Suspicious precision gap (69–72%) to determine whether class weighting or a confidence threshold on the Suspicious logit can reduce false positives without sacrificing recall.
- Explore Grad-CAM or attention weight visualisations to identify which of the 86 features and which flows within a 30-flow window the models weigh most heavily when predicting Suspicious traffic.